

AIForge — AI Assistant Platform

Project Description

AIForge is a ChatGPT-like AI assistant platform built using Spring Boot, React, and local Large Language Models through Ollama.

The platform supports:

- AI chat
- Streaming responses
- Authentication
- Chat history
- Markdown rendering
- File uploads
- Retrieval-Augmented Generation (RAG)
- AI tool calling
- Long-term memory

The goal of this project is to learn production-style AI engineering and full-stack development by building a real-world AI application from scratch.

Main Objectives

- Learn full-stack architecture
 - Learn AI integration with local LLMs
 - Understand streaming systems
 - Build scalable backend architecture
 - Learn vector databases and RAG
 - Build production-ready frontend systems
 - Prepare a strong placement-level portfolio project
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Target Users

- Developers
 - Students
 - Recruiters
 - AI enthusiasts
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MVP Features

Core Features

- AI chat
 - Streaming responses
 - JWT authentication
 - Chat history
 - Multiple conversations
 - Markdown rendering
 - File upload
 - RAG system
 - Tool calling
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Future Features

- Voice assistant
 - Multi-agent workflows
 - Web search
 - Team collaboration
 - Mobile app
 - AI image generation
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Tech Stack

Frontend

- React
- Vite
- TailwindCSS
- Zustand
- Axios

Backend

- Spring Boot
- Spring Security
- Spring WebFlux
- JWT

AI Stack

- Ollama
- Spring AI
- qwen3:8b

Databases

- PostgreSQL
- Qdrant

Deployment

- Docker
- Render/VPS

Final Expected Outcome

By completion the system should support:

- Real-time AI chat
- AI memory
- Document question answering
- AI tools
- Persistent conversations
- Production deployment
- Scalable architecture

Development Strategy

Every feature will be built using:

Backend → Testing → Frontend → Integration → Git Commit

No phase will be skipped.